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How is focal length defined for a two-lens system, separated by a distance d ?

Asked 8 years, 6 months ago Modified 4 months ago Viewed 93k times



I have found the formula for the effective [focal length](#) f of two thin lenses with focal lengths f_1 and f_2 separated by distance d to be

13

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}.$$



However, I can't seem to find how f is defined. Is it the distance from the first lens to the final focal point or the distance from the second lens to the final focal point? Or neither?



optics

definition

geometric-optics

lenses

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edited Sep 11, 2017 at 11:57



Qmechanic ♦

210k 48 575 2.3k

asked Apr 5, 2016 at 17:35



David Webb

253 1 2 6



The distance for the lens or lens set to converge an incoming parallel beam to a single point. For a diverging lens or set extend the diverging rays backwards until they converge on the optic axis. The distance from this point to the first deflection from parallel is the focal length, usually expressed as -ve for a divergence. – [Arif Burhan](#) May 18, 2016 at 19:47

2 Answers

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4



It is the distance from the image plane to the rear principal plane. You can find the location of this plane by projecting the image ray backwards through the system to where it crosses the projection of the object ray. This is sometimes also referred to as the effective focal length (v) of the system, and is true for both simple as well as complicated systems. The distance from the rear lens to the image plane is simply the back focal distance (v''). The difference between v'' and v can be found by the formula:

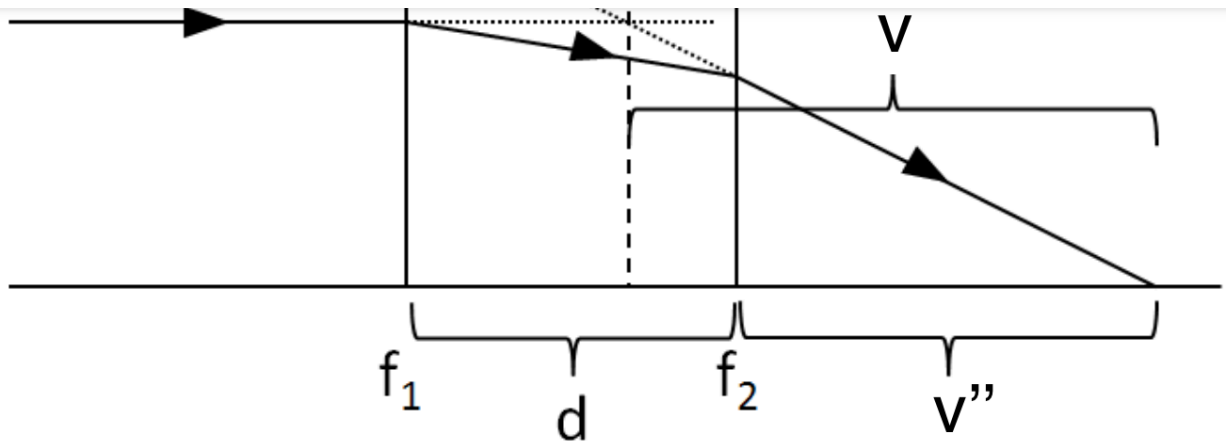
$$\delta = \frac{-d}{n} \frac{f}{f_1} = v'' - v$$



where $n = 1$ in air.

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edited Jun 8 at 8:52



Benjamin Wang

103 4

answered Apr 8, 2016 at 23:29



Andrew Stockham

159 7

▲ If I add the formula you gave for δ to the formula for BFD, $f_2^*(f_1-d)/(f_1+f_2-d)$, it does not appear to equal the formula for effective focal length. I derived δ myself and arrived at the same formula you gave so I assume it's correct, but I can't figure out why adding δ to BFD doesn't yield the formula for EFL that I have found in numerous sources. Do you know where the discrepancy lies? – David Webb Apr 11, 2016 at 3:02

▲ Your formula derives just fine. Take your time, sketch it all out, make sure you have the definitions correct, and post your work. I'll check it for errors if you'd like, but you have everything correct so far. – Andrew Stockham Apr 12, 2016 at 4:26

▲ I uploaded a photo of my work to photobucket:
i76.photobucket.com/albums/j23/dwebb1211/work_zpsqd05h4ej.png Please let me know if anything is wrong. – David Webb Apr 13, 2016 at 17:02

▲ You have the formula for δ as $-df_2/f_1$, it is $-dEFL/f_1$. – Andrew Stockham Apr 13, 2016 at 17:10

2 ▲ There are a lot of typos in this answer, and I found it hard to understand because of them.
– Steven Sagona Jul 9, 2021 at 12:37



I would rather just use the image of the previous answer, since it is really good, so please bear with me.

0



Consider that if we replace the system with the equivalent lens of focal length f (say).

We also assume that the left lens has a focal length f' and the right lens has a focal length f'' .



So, let's say the light ray intersects the equivalent lens at h when replaced, and the right lens(f'') at h' when not replaced (for the left lens also, the height is obviously h).



Observe, that h and h' form similar triangles post replacement whose base extends from the optics centre of their respective lens to the point where the image is formed.

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replacement.

$\Rightarrow x = fd/f'$ where x is the distance from the right lens(f'')

Sorry for not using latex

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answered Feb 2 at 12:49



[Sid N](#)

38 10



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